

Bio-tillage improves soil physical properties and maize growth in a compacted Vertisol by cover crops

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Abstract

Bio-tillage has recently been proposed as a measure to alleviate soil compaction through biopores created by cover crop roots. The objective of this study was to determine the effect of different cover crops on soil physical properties and the succeeding maize (*Zea mays* L.) growth in compacted soil. Four treatments, including no cover crop as a control (Con), alfalfa (*Medicago sativa* L.), oilseed rape (*Brassica napus* L.), and radish and hairy vetch mixture (*Raphanus sativus* L. and *Vicia villosa* Roth), were carried out under both compacted and noncompacted soil conditions. Soil physical properties, such as the volumetric soil water content (SWC), bulk density, saturated hydraulic conductivity (K_s) and air permeability at water potential of -60 hPa (K_{a60}), and maize root characteristics and yield were measured. The cover crops did not affect the soil bulk density but significantly decreased the SWC in both the compacted and noncompacted soils relative to the Con treatment. The alfalfa treatment presented significantly higher K_s in the noncompacted soil and K_{a60} in both the compacted and noncompacted soils than the Con treatment in the soil layer depth of 20–50 cm. The three cover crop treatments improved the maize root biomass density (173.2% for 2018 and 35.6% for 2019) and root length density (50.9% for 2018 and 51.8% for 2019) relative to the Con treatment in the soil layer depth of 10–70 cm in 2018 and soil layer depth of 10–50 cm in 2019 in the compacted soil rather than in the noncompacted soil. Compared with the Con treatment, the radish mixed with hairy vetch treatment in 2018 and the oilseed rape treatment in 2019 significantly enhanced the maize yield in the compacted soil. Our results suggest that alfalfa is the best crop for improving air permeability; however, the oilseed rape and mixture of radish and hairy vetch lead to better maize growth in the compacted soil. Bio-tillage using cover crops is effective in alleviating soil compaction.

Abbreviations: Alf, alfalfa; Con, control; K_{a60} , air permeability at water potential of -60 hPa; K_s , saturated hydraulic conductivity; Rap, oilseed rape; Rhv, radish mixed with hairy vetch; SWC, soil water content.

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1 | INTRODUCTION

Soil compaction has been a growing major concern for agronomists and farmers because increasingly heavy agricultural machinery is being widely used in field management (Graves et al., 2015; Hamza & Anderson, 2005; Schjønning et al., 2015). Soil compaction results in high penetration resistance and low oxygen concentrations in soil air due to the collapse of void space, and both of these changes are adverse for root growth and water and nutrient uptake for crops (Ishaq et al., 2001; Nosalewicz & Lipiec, 2014; Xiong et al., 2020). Consequently, crop productivity has been greatly reduced in compacted soil (Blanco-Canqui & Lal, 2008; Bogunovic et al., 2018; Dauda & Samari, 2002). For example, Correa et al. (2019) reviewed many studies that investigated several crops and soils in numerous countries and reported that crop yield losses by compaction ranged from 5 to 75%. Moreover, the negative effect of subsoil compaction caused by heavy vehicles on crop productivity can last over 5 yr (Chamen et al., 2015; Håkansson & Reeder, 1994). Therefore, it is urgent to develop strategies to restore crop production in compacted soils to safeguard food security globally.

Mechanical loosening, such as subsoiling, is mostly applied to disrupt compacted soil layers in the subsoil (Calonego & Rosolem, 2010; Hamza & Anderson, 2005). However, the benefits of subsoiling are short lived because loosened soil is particularly susceptible to recompaction due to subsequent machinery traffic and repeated rainfall events (Busscher et al., 2002; Calonego et al., 2017). Incorporation of cover crops with deep and strong root systems that have the ability to penetrate compacted soil into the crop rotation has been suggested as a method of mitigating soil compaction (Chen & Weil, 2011; Cresswell & Kirkegaard, 1995; Kautz, 2014). The roots of cover crops can create voids by penetrating the compacted soil and leaving biopores when they are decomposed, which is usually termed “bio-tillage” (Elkins, 1985; Han et al., 2015; Zhang & Peng, 2021). The biopores could improve soil water or air permeability, provide preferential space with lower soil strength and higher oxygen concentrations for crop root growth, and consequently enhance crop performance in compacted soil (Athmann et al., 2013; Colombi et al., 2017; Zhang et al., 2019). Colombi et al. (2017) found that the root dry weight for both wheat (*Triticum aestivum* L.) and soybean [*Glycine max* (L.) Merr.] was approximately 40% higher in compacted soil with biopores than in such soil without biopores. Peth et al. (2014) observed that cover crop treatments significantly increased air permeability in compacted soil relative to a fallow treatment due to the presence of root channels. Furthermore, they also speculated that these root channels enhanced maize (*Zea mays* L.) root penetration of compacted soil and facilitated access to subsoil water (Chen & Weil, 2011). However, to date, few studies have evaluated the effects of cover crop on alleviating soil compaction.

Core Ideas

- Cover crops significantly reduced soil water content but did not alter bulk density.
- Alfalfa was the best crop for improving air permeability in compacted soil.
- Cover crops improved the maize root characteristics and yield in compacted soil.
- Oilseed rape and a mixture of radish and hairy vetch had better effects on maize growth.

Identifying cover crop species that are good at moderating the adverse effects of soil compaction is of great importance for the use of bio-tillage. Garcia et al. (2013) noted that fibrous-rooted sorghum [*Sorghum bicolor* (L.) Moench]–sudangrass [*Sorghum bicolor* (L.) Moench ssp. *drummondii* (Nees ex Steud.) de Wet & Harlan] and pearl millet (*Pennisetum glaucum* (L.) R. Br.) increased soil porosity (total, macro, and micro) and aggregate stability and decreased bulk density compared with fallow due to their extensive root systems. However, tap rooted cover crops with larger diameters usually present more potential in drilling strong subsoil than fibrous-rooted crops (Chen & Weil, 2010; Han et al., 2015). Chen and Weil (2010) indicated that the ability to penetrate compacted soil for cover crops was in the order of forage radish (*Raphanus sativus* L.) > rapeseed (*Brassica napus* L.) > rye (*Secale cereale* L.). Abdollahi et al. (2014) also revealed that cover cropping with fodder radish considerably increased continuous macropores and improved air permeability at -3 kPa matric potential in the topsoil (4–16 cm) and plow pan (18–27 cm). However, it was also reported that radish did not perform as well as chicory (*Cichorium intybus* L.) and alfalfa (*Medicago sativa* L.) in improving the pore structure of compacted soil, although all of these plants have taproots (Pulido-Moncada et al., 2020).

Different cover crop species also result in variations in the performance of succeeding crops. Han et al. (2016) found that the diameter of barley (*Hordeum vulgare* L.) roots grown after tap rooted chicory shifted toward larger size classes compared with that of barley roots grown after fibrous-rooted tall fescue (*Festuca arundinacea* Schreb.) because chicory created more large-sized biopores than the tall fescue. Williams and Weil (2004) documented that the soybean yield following radish was significantly lower than that following rye at one site but slightly higher at another site, which likely resulted from different severities of compaction and precipitation at the two sites. In general, the species of cover crops that are the most effective in improving crop growth in compacted soils have not been clarified and need further investigation.

Both brassicas and alfalfa have thick and strong roots and have been reported to be effective in mitigating soil compaction (Chen & Weil, 2011; Han et al., 2015; Pulido-Moncada et al., 2020). It has been suggested that a mixture of brassicas and legumes is more beneficial for succeeding crops than individual crops (Chu et al., 2017). In this study, four treatments, including no cover crop as a control (Con), alfalfa (Alf), oilseed rape (Rap), and radish mixed with hairy vetch (Rhv) were carried out in compacted and noncompacted soils. Our objective was to compare the effect of different cover crops on soil physical properties and maize growth in compacted soil. The hypotheses were that a mixture of radish and hairy vetch performed better than alfalfa and oilseed rape at ameliorating the (a) physical properties and (b) maize growth in compacted soil.

2 | MATERIALS AND METHODS

2.1 | Experimental site and design

The experimental site was located at Longkang Farm, Anhui Province, China (33°32' N, 115°59' E, and 23 m in elevation), with an annual precipitation of 924 mm and a mean annual temperature of 16.2 °C. The soil is a clay loam (sand, 8%; silt, 54.1%; and clay, 37.9%) and classified as a Vertisol according to the USDA Soil Taxonomy. Before the experiment, the cropping system was winter wheat (October–June)–summer maize (June–September). The soil was tilled with a rotary tiller to a depth of 12–15 cm for wheat sowing, whereas maize was sown directly without any tillage after wheat harvest in 2015 and 2016.

The experiment was established in November 2017 with a summer maize–winter cover crop rotation. Two factors were involved in this trial. The first factor was soil compaction, including compacted and noncompacted levels, and the second factor was cover crop, including no cover crop as a Con, Alf, Rap, and Rhv. Thus, this experiment produced a total of eight treatments (two compactations $\times$ four cover crops). Each treatment was performed in triplicate. There were 24 plots in total, with a size of 7 $\times$ 10 m for each plot. The same plot was used for the same treatment every year. Before the soil treatments, the soil water content was 0.31 cm³ cm⁻³, 0.33 cm³ cm⁻³, 0.34 cm³ cm⁻³, and 0.36 cm³ cm⁻³, and the bulk density was 1.40 g cm⁻³, 1.58 g cm⁻³, 1.58 g cm⁻³, and 1.48 g cm⁻³ at soil depths of 0–10 cm, 10–20 cm, 20–30 cm, and 30–50 cm, respectively. The compaction treatment was prepared by a 7,400-kg tractor passing three times track by track to ensure that the entire plot area was compacted; the noncompacted treatment was subsoiled to a 35-cm soil depth with a 3,920-kg tractor. Compaction and subsoiling in the respective plots were both carried out only once at the beginning of the experiment. Then, the field was tilled using a rotary tiller to

a depth of 5–10 cm to prepare an appropriate seedbed for all the treatments. The cover crops were planted by broadcasting with seeding rates of 21.4 kg ha⁻¹ for Alf and Rap, and 42.8 kg ha⁻¹ for Rhv (1:1) on 3 Nov. 2017 and 16 Sept. 2018, respectively. Nitrogen and phosphate fertilizers (60 kg N ha⁻¹ and 45 kg P₂O₅ ha⁻¹) were applied before rotary tillage in 2017, and only N fertilizer (30 kg N ha⁻¹) was applied in the seedling period of cover crops in 2018 to ensure good growth of these cover crops. The cover crops were harvested manually with sickles in the following May and the aboveground biomass was removed from the field to minimize soil mulch effects on soil properties.

The maize seeds were sown manually with a space of 30 $\times$ 60 cm after rotary tillage to a depth of 5–10 cm on 2 June 2018 and 17 May 2019. Basal fertilizers, including 100 kg N ha⁻¹, 60 kg P₂O₅ ha⁻¹, and 90 kg K₂O ha⁻¹ were applied before rotary tillage, and an additional 80 kg N ha⁻¹ was applied at the 8–10 leaf stage of maize growth. The maize was harvested manually in September, and the straw was removed from the field in both years.

2.2 | Soil sampling and measurements

In each plot, a polyvinyl chloride access tube (length 1 m, diameter 0.05 m) was vertically installed to a 0.8-m soil depth in December 2018. The bottom of the tube was sealed with a rubber lid to prevent possible effects of groundwater on water content measurement. A portable probe (time domain reflectometry, IMKO) was used to measure the volumetric soil water content (SWC) in four soil layers (0–20 cm, 20–40 cm, 40–60 cm, and 60–80 cm). The SWC was determined approximately once per week in the cover crop season and once every 2 wk in the maize season. Soil penetration resistance was measured at 10 randomly selected locations per plot using a cone penetrometer (Rimik, CP40II) with 12.8-mm diameter and 30° apex angle. The soil penetration resistance was recorded every 2.5 cm to a depth of 65 cm for each measurement. In 2018, soil penetration resistance was measured immediately after harvesting the cover crops. In 2019, soil penetration resistance was measured after a heavy rainfall in the maize seedling stage because the soil was too hard to measure this parameter after harvesting the cover crops due to the low SWC.

In June 2019, intact soil cores were sampled vertically with PVC cylinders (86 cm³, 4.68 cm in diameter and 5.00 cm in height) in soil depths of 0–10 cm, 20–30 cm, and 30–50 cm in each plot. The soil cores were wrapped with plastic film and transported to the laboratory carefully to avoid soil disturbance. The samples were slowly saturated in a sandbox with the bottom 3 cm immersed in water. Then, the water potential of the sandbox was adjusted to -60 hPa to drain the water in the soil cores. After reaching a balance, the air

permeability (K_a) was measured with the steady state method under a constant air pressure difference of 30 cm water pressure head (Zhang et al., 2019). The air permeability of the soil cores determined at a water potential of -60 hPa was indicated as K_{a60} .

After measuring the air permeability, the soil cores were saturated again in the sandbox. Then, the saturated hydraulic conductivity (K_s) was measured using the falling head method (Reynolds et al., 2002). Afterward, the soil cores were weighed after oven-drying at 105°C for 48 h to determine the soil bulk density.

2.3 | Maize root characteristics and yield

In each plot, soil cores with 10-cm diameter were sampled at locations 10 cm away from a random maize stem in both the row and perpendicular to row directions on 24 Aug. 2018 and 4 Sept. 2019, respectively. The cores were taken every 10 cm from the soil surface to a depth of 70 cm. Root samples were washed out from each core and then scanned to determine root length density using the WinRhizo software (Regent Instruments). Then, the root samples were oven-dried at 80°C to calculate the root biomass density. The maize yield was obtained by harvesting all the maize seeds in each plot.

2.4 | Statistics

The soil compaction, cover cropping and their interactive effects on the soil properties, and maize root characteristics and yield were assessed using ANOVA at the significance level of $P < .05$. In the post hoc multiple comparisons, the differences in these parameters among treatments were tested with the LSD method at the $P < .05$ level. Saturated hydraulic conductivity, air permeability, root biomass density, and root length density were ln-transformed for normality before the ANOVA. Spearman's correlation analysis was performed to test the relationships between the maize root characteristics and yield and the soil physical properties. All the above analyses were conducted with the statistical software package IBM SPSS Statistics 19.

3 | RESULTS

3.1 | Effect of cover crops on the soil physical properties

Soil compaction significantly increased the SWC of the topsoil (0–20 cm) throughout the whole cover crop season and at the early stage of the maize season ($P < .05$; Figure 1). In the cover crop season, the SWC showed a little temporal variation in the topsoil (0–20 cm), whereas it was relatively stable in the

soil layer depth of 20–80 cm before the end of February 2019. The SWC sharply decreased in the soil layer depth of 0–40 cm in April 2019, especially for the treatments with cover crops. The SWC was significantly lower in all cover crop treatments compared with the Con treatment for both the compacted and noncompacted treatments in the topsoil (0–20 cm) and for the noncompacted treatment in soil layer depth of 20–40 cm in April 2019. At the soil layer depth of 40–60 cm, the SWC was significantly lower in the Alf and Rhv treatments than in the Con treatment for the noncompacted plots ($P < .05$), whereas the difference in SWC was minor among the four treatments for the compacted treatment in April 2019. On July 3, the SWC was still significantly lower in the Alf treatment compared with the Con treatment in the soil layer depth of 20–60 cm for the noncompacted treatment and in soil layer depth of 0–60 cm for the compacted treatment ($P < .05$).

Soil penetration resistance was significantly lower in the noncompacted treatment than in the compacted treatment at 0–25 cm and 35–52.5 cm in 2018, and at 5–25 cm, 42.5–45 cm, and 57.5–65 cm in 2019 ($P < .05$; Figure 2). The difference in soil penetration resistance between the compacted and noncompacted soils at the soil layer depth of 0–22.5 cm was smaller in 2019 than in 2018. The soil layer depth of 10–30 cm in the Alf treatment, and 7.5–17.5 cm and 20–25 cm in the Rhv treatment presented significantly higher soil penetration resistance than the Con treatment for the compacted soil in 2018 ($P < .05$). However, the Con treatment showed significantly greater soil penetration resistance than the Rhv, Rap, and Alf treatments at soil depths of 5–15 cm, 7.5–12.5 cm, and 10–12.5 cm, respectively ($P < .05$), for the noncompacted soil in 2019. No significant effect of cover crops on soil penetration resistance was observed in the noncompacted soil in 2018 or in the compacted soil in 2019.

Soil compaction significantly influenced the soil bulk density ($P < .05$; Table 1), but cover cropping did not influence the soil bulk density in any of the three soil layers. Multiple comparisons showed that in the soil layer depth of 30–50 cm, the bulk density was significantly higher in the Alf treatment than in the Con treatment under compacted conditions ($P < .05$; Figure 3).

Soil compaction significantly reduced the K_s in the top 30-cm soil layers, whereas the cover crops significantly increased K_s in the soil layer depth of 20–50 cm ($P < .05$; Table 1; Figure 4; Supplemental Table S1). Under noncompacted conditions, the Alf treatment had a significantly higher K_s value than the other treatments in the soil layer depth of 20–50 cm, except for the Rhv treatment in the soil layer depth of 30–50 cm ($P < .05$). The K_s in the Rhv treatment in the soil layer depth of 30–50 cm was significantly higher than that in the Con treatment for the noncompacted soil ($P < .05$). However, there was no significant difference in the K_s between the Con and other cover crop treatments under compacted conditions.

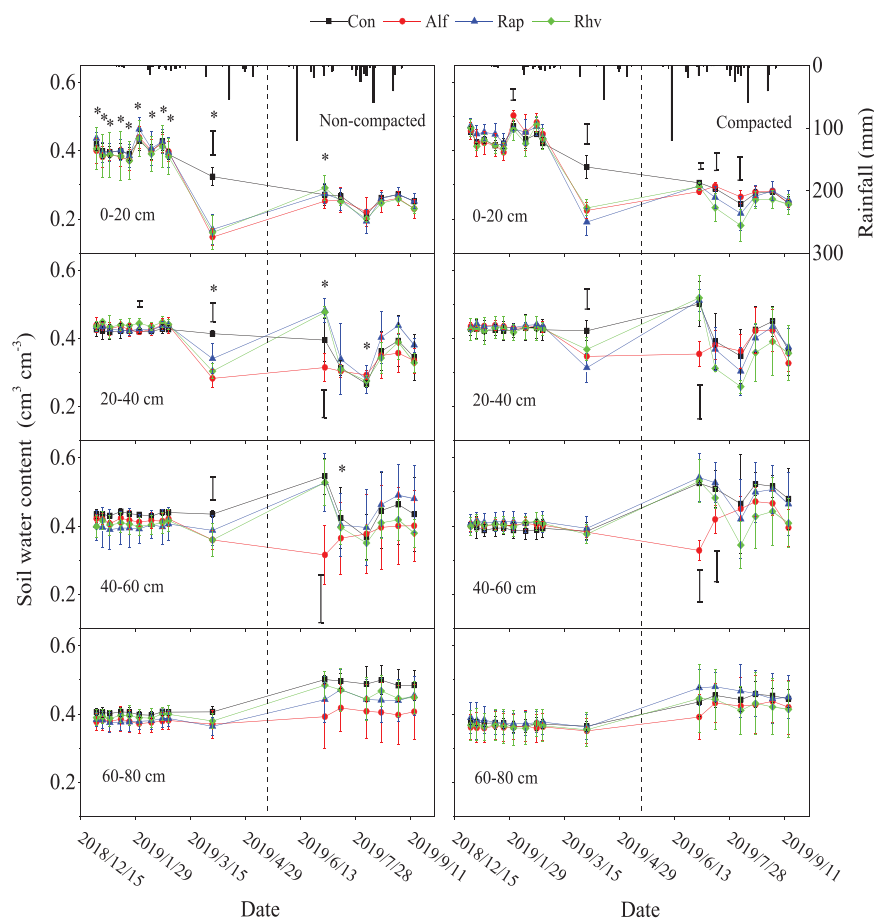


FIGURE 1 Temporal changes in soil water content in different soil layers of the soil compaction and cover crop treatments. The vertical bars indicate significant LSD values at $P < .05$ among the four cover crop treatments at each compaction level. The vertical dashed line indicates the day for maize sowing. Alf, alfalfa; Con, control; Rap, oilseed rape; Rhv, radish mixed with hairy vetch. *Significant difference ($P < .05$) between the compacted and noncompact treatments

Soil compaction, cover cropping, and their interaction had significant effects on the K_{a60} in soil layer depth of 20–50 cm ($P < .05$; Table 1). Regardless of compaction or not, the Alf treatment, but not the Rap and Rhv treatments, presented a significantly higher K_{a60} than the Con treatment in both the soil layer depths of 20–30 cm and 30–50 cm ($P < .05$; Figure 5; Supplemental Table S1).

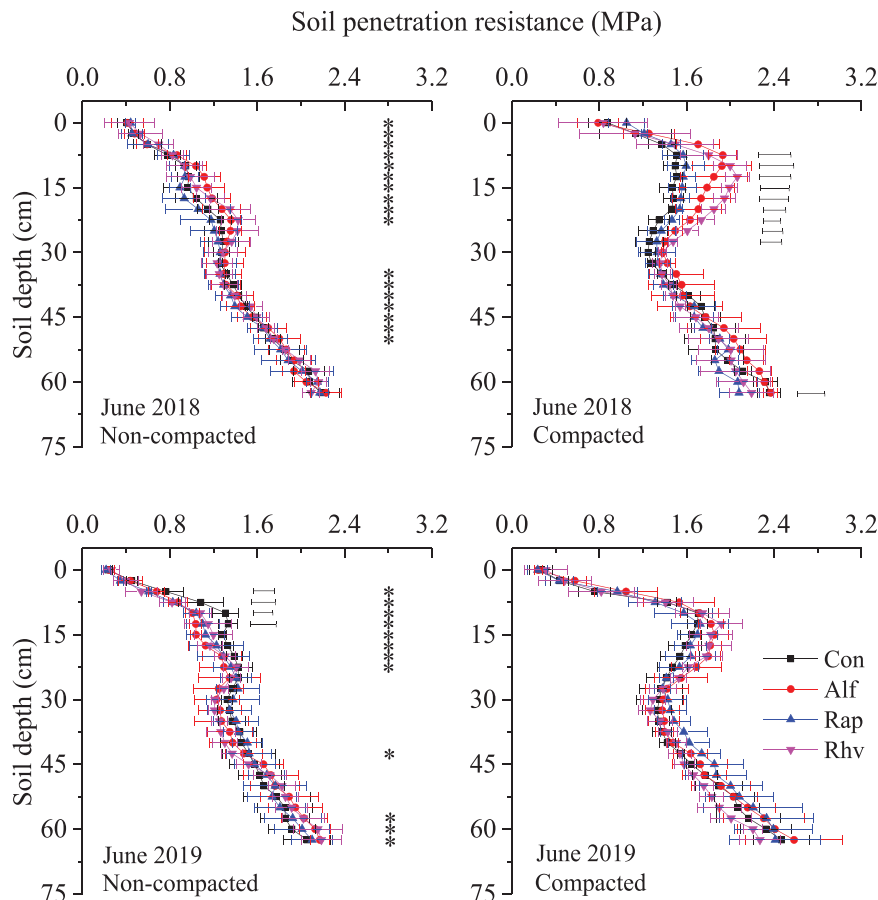
3.2 | Effect of cover crops on maize root characteristics and yield

Soil compaction had a significant effect on the root length density in the soil layer depth of 60–70 cm in 2018 and in the soil layer depth of 0–10 cm in 2019 ($P < .05$, Figures 6 and 7; Supplemental Tables S2 and S3). In the noncompact treatment, the root biomass and length densities at all the soil layers in the Con treatment were similar to those in the other three cover crop treatments in 2018 and 2019. However, in the compacted soil, the three cover crop treatments generally increased the maize root biomass density (173.2% for 2018 and 35.6% for 2019) and root length density (50.9% for 2018 and 51.8% for 2019) relative to the Con treatment in the soil layer depths of 10–70 cm in 2018 and 10–50 cm in 2019. Moreover, the root biomass density was significantly greater

in the Rap treatment than in the Con treatment for the soil layer depths of 10–20 cm and 40–50 cm ($P < .05$), and the root length density was significantly greater in the Rap and Alf treatments than in the Con treatment for the soil layer depth of 50–60 cm in 2018 under compacted soil conditions ($P < .05$). The Con treatment had significantly lower root length density than the Alf and Rhv treatments in the soil layer depth of 10–20 cm and the Rhv treatment in the soil layer depth of 30–40 cm in 2019 under compacted soil conditions ($P < .05$).

The maize yield was 48.4% lower on average in 2018 than in 2019, presenting a large variation between the 2 yr (Figure 8). Soil compaction significantly decreased the maize yield in 2018 ($P < .05$) but did not decrease the maize yield in 2019 regardless of the preceding cover crop that was planted. The Rhv and Rap treatments increased the maize yield more under compacted conditions (8.65–65.3%) than under noncompact conditions (7.58–17.7%) relative to the Con treatment in the 2 yr. Under compacted conditions, the Rhv treatment, but not the Alf and Rap treatments, significantly enhanced the maize yield (by 65.3%) compared with the Con treatment ($P < .05$), although the latter two treatments also exhibited 40.2 and 42.4% higher maize yields in 2018, respectively. In 2019, the Rap treatment significantly enhanced the maize yield by 19.9% relative to the Con treatment in the compacted soil ($P < .05$). However, the maize

FIGURE 2 Soil penetration resistance in different soil layers of the soil compaction and cover crop treatments in 2018 and 2019. The horizontal bars indicate significant LSD values at $P < .05$ among the four cover crop treatments at each compaction level. Alf, alfalfa; Con, control; Rap, oilseed rape; Rhv, radish mixed with hairy vetch. *Significant difference ($P < .05$) between the compacted and noncompacted treatments



yield in the Alf treatment decreased by 7.05 and 13.6% relative to the Con treatment in the compacted and noncompacted conditions, respectively.

4 | DISCUSSION

4.1 | Bio-tillage by cover crops modifying soil physical properties

The three cover crops significantly decreased the SWC in the soil layer depth of 0–20 cm compared with the Con treatment in the spring of 2019, in both the compacted and noncompacted treatments (Figure 1). A similar result was observed by Nielsen et al. (2015), who found that cover crops planted in single-species or multi-species mixtures decreased the SWC in the soil layer depth of 0–60 cm in both Akron, CO and Sidney, NE in the United States, where the annual precipitation was 416 and 427 mm, respectively. The result was likely associated with the much greater water use of cover crops than evaporative water loss from the fallow in the study area (Nielsen et al., 2015; Unger & Vigil, 1998). However, Blanco-Canqui et al. (2015) stated that cover crops often enhanced the SWC in regions with high precipitation inputs (>800 mm). At this study site, the rainfall in 2019 was only 631 mm, with

111 mm in the spring (March–May), although the average annual rainfall was 942 mm (Supplemental Figure S1). The difference in SWC between the Con treatment and cover crop treatments in the soil layer depth of 40–60 cm was more pronounced in the noncompacted plots than in the compacted plots in April 2019 in our study (Figure 1). This result is consistent with that of Nosalewicz and Lipiec (2014), who reported that wheat water uptake was much higher in low and moderately compacted soil than in heavily compacted soil. This result can be attributed to soil compaction reducing the volume of larger pores but enhancing the volume of smaller water retention pores, which consequently inhibits water drainage and evaporation (Alaoui et al., 2011; Tian et al., 2018). More importantly, the aboveground biomass of cover crops in the noncompacted plots was much higher than that in the compacted plots (Yan et al., 2020), resulting in greater transpiration and water loss from deeper soil in the noncompacted plots (Nosalewicz & Lipiec, 2014).

In this study, after the experiment was conducted for almost 2 yr, the bulk density was still significantly greater in the compacted plots; however, the cover crop treatments had little effect on the bulk density in all three soil layers (Figure 3). In two previous meta-analysis studies, a negligible effect of cover crops on bulk density was also reported (Alvarez et al., 2017; Blanco-Canqui & Ruis, 2020). The change in bulk

TABLE 1 Analysis of variance on the bulk density, saturated hydraulic conductivity, and air permeability at different soil layers affected by soil compaction, cover crop, and their interaction

Soil depth	Variation source	Bulk density	$\ln(K_s)$	$\ln(K_{a60})$
0–10 cm	Compaction	0.002	0.028	0.270
	Cover crop	0.178	0.192	0.703
	Compaction–cover crop	0.205	0.551	0.143
20–30 cm	Compaction	0.000	0.000	0.000
	Cover crop	0.631	0.004	0.001
	Compaction–cover crop	0.682	0.001	0.019
30–50 cm	Compaction	0.004	0.080	0.000
	Cover crop	0.144	0.042	0.000
	Compaction–cover crop	0.103	0.363	0.000

Note. The bold numbers indicate significance at $P < .05$. K_{a60} , air permeability; K_s , hydraulic conductivity; Ln, natural logarithm.

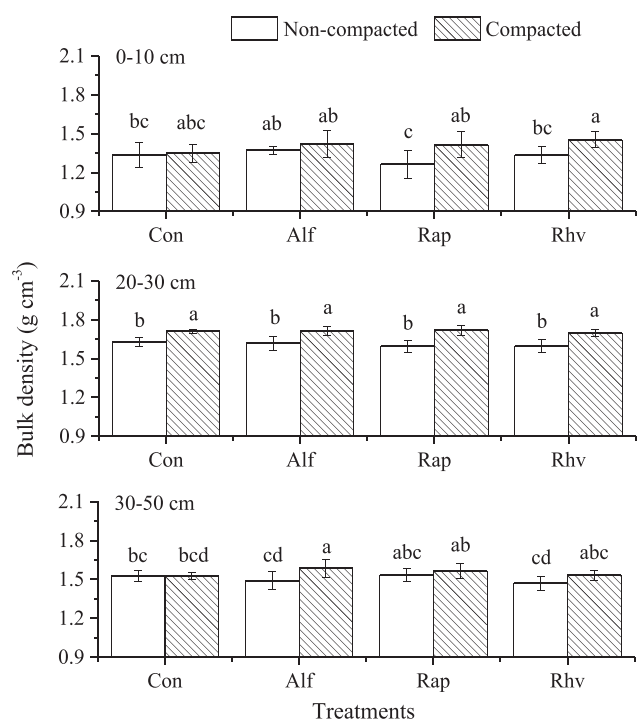


FIGURE 3 Bulk density in different soil layers of the soil compaction and cover crop treatments. Different letters indicate significant differences at $P < .05$ among the eight treatments. Alf, alfalfa; Con, control; Rap, oilseed rape; Rhv, radish mixed with hairy vetch

density usually results from vertical movement of soil particles (Keller et al., 2021). The cover crop roots only exert radial pressure on the surrounding soil and result in compression of the soil at approximately 1 mm around the roots (Angers & Caron, 1998; Koebnick et al., 2019), which does not necessarily induce vertical movement of soil particles and subsequent changes in bulk density. Furthermore, the effect of cover crops on bulk density is likely dependent on cover cropping duration (Blanco-Canqui et al., 2015). As revealed in most

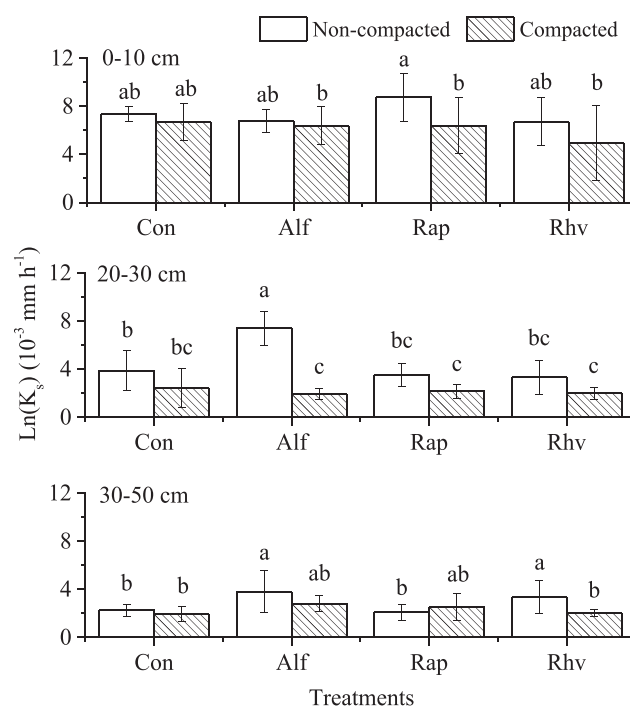


FIGURE 4 Saturated soil hydraulic conductivity in different soil layers of the soil compaction and cover crop treatments. Different letters indicate significant differences at $P < .05$ among the eight treatments. Alf, alfalfa; Con, control; K_s , hydraulic conductivity; Ln, natural logarithm; Rap, oilseed rape; Rhv, radish mixed with hairy vetch

studies, cover cropping can increase the SOC over the long term, particularly due to the input of belowground biomass, which leads to improved soil aggregation and decreased soil bulk density (Blanco-Canqui & Ruis, 2020; Or et al., 2021). Thus, lower bulk density after cover cropping was observed in a 15-yr experiment (Blanco-Canqui et al., 2011) and a 12-yr experiment (Steele et al., 2012), which were much longer than our 2-yr experiment.

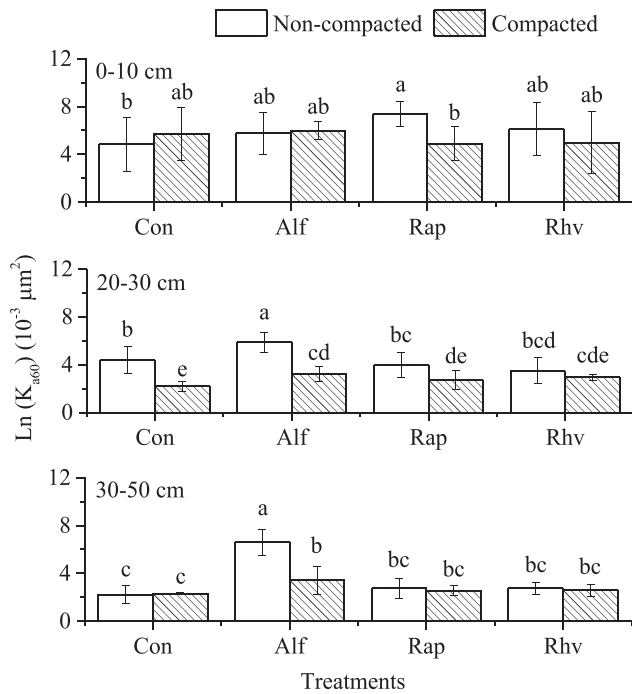
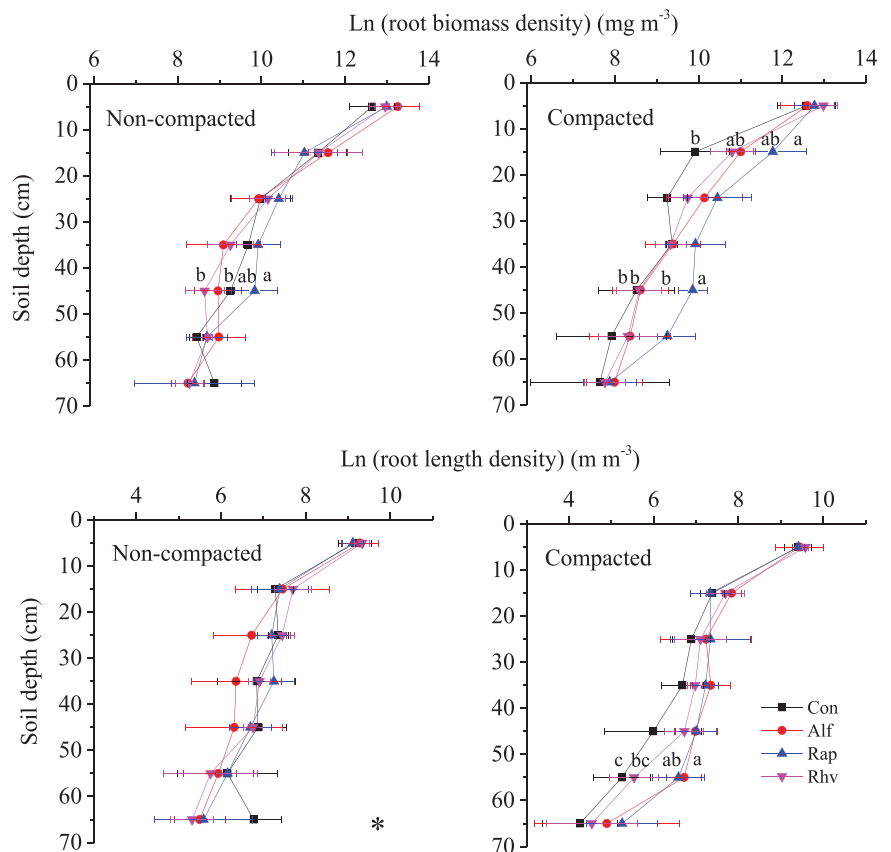


FIGURE 5 Air permeability in different soil layers of the soil compaction and cover cropping treatments. Different letters indicate significant differences at $P < .05$ among the eight treatments. Alf, alfalfa; Con, control; K_{a60} , air permeability; Ln, natural logarithm; Rap, oilseed rape; Rhv, radish mixed with hairy vetch

FIGURE 6 Maize root biomass density and root length density in the soil compaction and cover crop treatments in 2018. Different letters indicate significant differences at $P < .05$ among the four cover crop treatments at each compaction level. Alf, alfalfa; Con, control; Ln, natural logarithm; Rap, oilseed rape; Rhv, radish mixed with hairy vetch. *Significant difference ($P < .05$) between the compacted and noncompact treatments



We observed that the Alf and Rhv treatments significantly increased soil penetration resistance at soil layer depth of 10–30 cm under compacted conditions in 2018, and the Con treatment presented significantly higher penetration resistance than the other three treatments at soil layer depth of approximately 5–15 cm in the noncompact plots in 2019 (Figure 2). It is well known that soil penetration resistance is determined by the bulk density and water content (Gao, Whalley, et al., 2016). Thus, we attributed the significant difference to the variation in SWC between the Con treatment and cover crop treatments because the cover crops did not have an observed effect on the soil bulk density, even after 2 yr of cover cropping (Figure 3). Ruis et al. (2020) also attributed the difference in soil penetration resistance to the variation in SWC in a cover crop experiment. The response of soil penetration resistance to cover crops varied between the compacted and noncompact treatments in both years (Figure 2). This result was likely associated with the different soil water contents between the two compaction levels resulting from diverse hydraulic properties, such as the saturated hydraulic conductivity and water retention (Figures 1 and 4).

Cover cropping had a significant effect on the K_s and K_{a60} in both the soil layer depths of 20–30 cm and 30–50 cm in this study (Table 1), indicating that air permeability and saturated hydraulic conductivity were much more sensitive to cover cropping than bulk density. Keller et al. (2021)

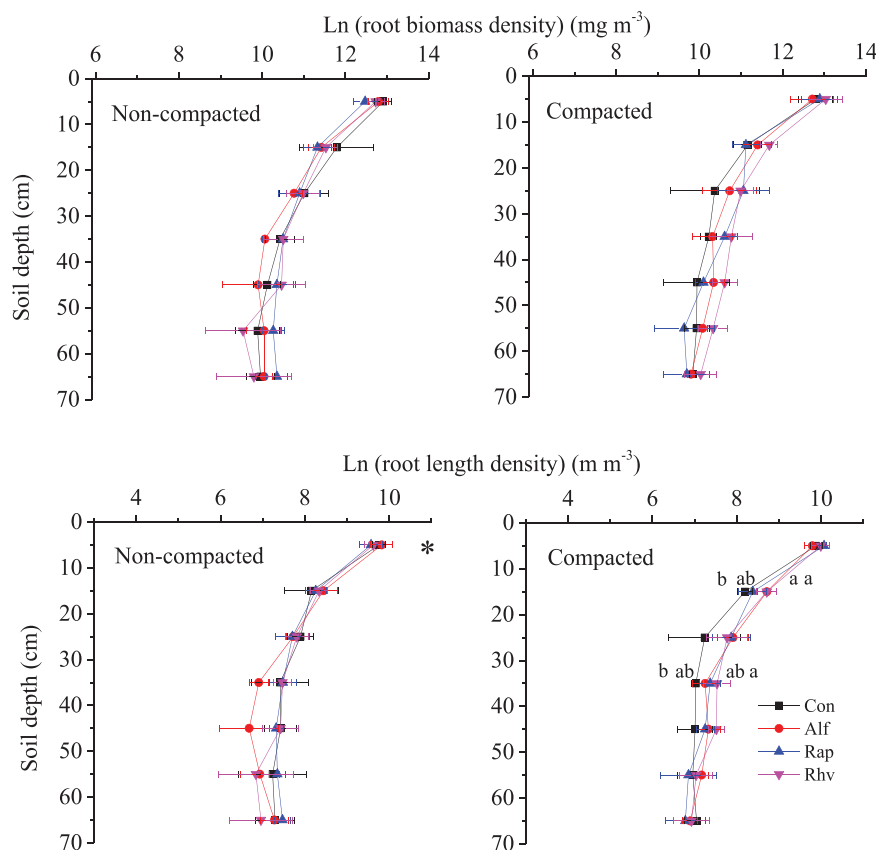


FIGURE 7 Maize root biomass density and root length density in the soil compaction and cover crop treatments in 2019. Different letters indicate significant differences at $P < .05$ among the four cover crop treatments at each compaction level. Alf, alfalfa; Con, control; Ln, natural logarithm; Rap, oilseed rape; Rhv, radish mixed with hairy vetch. *Significant difference ($P < .05$) between the compacted and noncompacted treatments

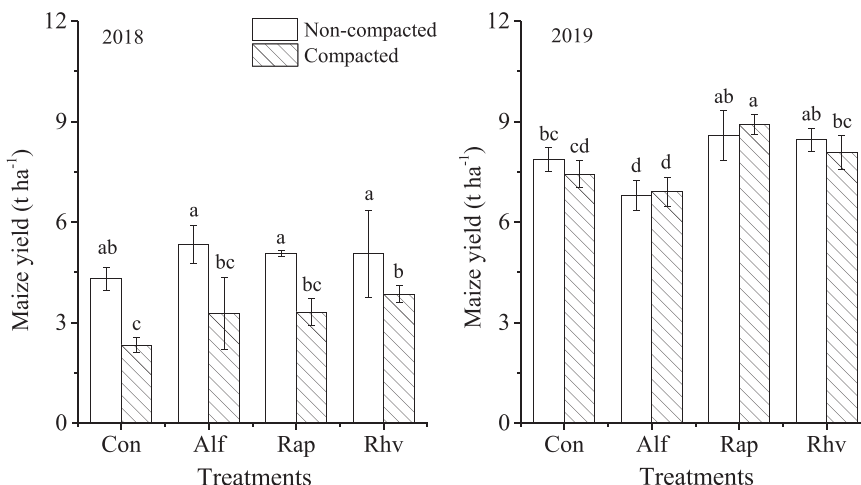
suggested that air permeability and saturated hydraulic conductivity in compacted soil can be quickly improved, because they are determined by highly continuous biopores that can be easily formed by the roots of cover crops (Zhang et al., 2019). In this study, the strong taproots of the cover crops were able to create new continuous biopores or unblock the existing biopores, consequently enhancing the K_s and K_{a60} . In a previous study, Peth et al. (2014) revealed that both radish and rape significantly increased the air permeability at a soil layer depth of 0–12 cm in highly compacted plots. Abdollahi et al. (2014) also reported significantly higher air permeability at soil layer depths of 4–16 cm and 18–27 cm after radish cropping. However, in the current study, the effect of Rhv treatment on the K_{a60} was minor (Figure 5). The Alf treatment was most effective at improving the K_s and K_{a60} in the soil layer depths of 20–30 cm and 30–50 cm for both compaction levels but not for the K_s at 20–30 cm under compacted plots (Figures 4 and 5). The result was similar to that of previous studies (Kautz et al., 2010; Pulido-Moncada et al., 2020) but did not support our hypothesis that radish mixed with hairy vetch performed better at improving soil physical properties. Angers and Caron (1998) suggested that alfalfa has a large diameter taproot that is long and almost straight and thus effective in improving macropore flow. We likewise found that the root volume of alfalfa was greater than that of the radish and hairy vetch mixture and rape at a soil layer depth of 10–50 cm for both noncompacted and compacted treatments in a previous

study (Yan et al., 2020), likely resulting in larger macroporosity and greater conductivity of air and water in the Alf treatment. We also found that the Alf treatment effects on water and air conductivities were more prominent in the noncompacted treatment than in the compacted treatment (Figures 4 and 5). In a previous study, the compacted treatment decreased the root volume of alfalfa by 64% at a soil layer depth of 0–70 cm in 2018 relative to that of the noncompacted treatment (Yan et al., 2020). The lower root volume was less effective in improving the soil pore structure and consequently the K_s and K_{a60} (Kochiieru et al., 2020). Another explanation was that crop roots prefer to use existing macropores rather than penetrate the soil matrix when growing in compacted soil with high strength (Atkinson et al., 2020; Gao, Hodgkinson, et al., 2016), resulting in a smaller number of new biopores in the compacted soil.

4.2 | Bio-tillage by cover crops improving maize growth

We observed that the root biomass density and root length density in the Con treatment were generally lower than those in the three cover crop treatments in the soil layer depths of 10–70 cm for 2018 and 10–50 cm for 2019 in the compacted plots but not in the noncompacted plots (Figures 6 and 7). The results are consistent with that of Bertollo et al. (2021), who

FIGURE 8 Maize yield in the soil compaction and cover crop treatments in 2018 and 2019. Different letters indicate significant differences at $P < .05$ among the eight treatments. Alf, alfalfa; Con, control; Rap, oilseed rape; Rhv, radish mixed with hairy vetch



noted that the preceding cover crop increased the soybean root biomass and length density, mainly in the deeper layers of the compacted treatments. Chen and Weil (2011) also revealed that the effect of cover crops on improving maize root penetration in subsoil was more pronounced at the high compaction level than at the medium compaction level. Although the soil pore structure may be less improved in the compacted treatment than in the noncompacted treatment after the cultivation of cover crops, as indicated by the K_{a60} (Figure 5), the biopores derived from cover crops played a more important role in promoting maize root growth in the compacted treatment with higher soil strength. It has been reported that more crop roots tend to grow in biopores in compacted soil than in loose soil, where soil strength is not a constraint for root growth (Atkinson et al., 2020). Moreover, only a very small portion of soil macropores, usually below 20%, is used for root growth, especially in denser soil (White & Kirkegaard, 2010). The root channels created by cover crops in compacted soils can improve the succeeding maize root growth by certain mechanisms, such as providing favorable zones with low mechanical resistance, high oxygen concentrations, and high nutrient contents, and facilitating root uptake of water and nutrients in the deeper bulk soil by bypassing compacted soil layers through biopores (Atkinson et al., 2020; Colombi et al., 2017; Kautz, 2014; Rasse & Smucker, 1998; Williams & Weil, 2004). In this study, the root biomass density and length density in the compacted treatment were similar to those in the noncompacted treatment for the soil layer depth of 10–60 cm, although the compacted treatment had significantly higher bulk density in the soil layer depth of 20–50 cm and penetration resistance in soil layer depth of 10–25 cm (Figures 2, 3, 6, and 7). This result further confirmed that the biopores in the compacted treatment might improve maize root growth.

Soil compaction seemed to increase the root length density at the soil layer depth 0–10 cm (Figures 6 and 7) because the root length density in this layer was positively correlated with the soil bulk density in the soil layer depth of 0–30 cm in 2018

and 2019 and soil penetration resistance in the soil layer depth of 0–50 cm in 2018 (Supplemental Table S4). Rosolem et al. (2002) also reported that the root length density increased with increasing penetration resistance for some crop species because the roots tend to proliferate horizontally with many fine lateral roots when encountering vertical resistance, thus leading to higher root length density. On the other hand, soil with a lower bulk density and improved air or water conductivities was beneficial for root penetration to deeper soil layers and did not require the proliferation of many fine lateral roots (Bengough et al., 2006).

The maize yield in 2018 was much lower than that in 2019 for all the treatments (Figure 8). The result can be attributed to different rainfall levels in the 2 yr. The rainfall in the maize growing season (June to September) was 1.8 times larger in 2018 (703 mm) than in 2019 (391 mm), resulting in waterlogging in 2018. Moreover, the low elevation of the experimental site and poor drainage capacity of the Vertisol (Erkossa et al., 2014) aggravated the effect of waterlogging on the maize yield in 2018. The maize yield was significantly lower in the compacted treatment than in the noncompacted treatment in 2018, but statistical difference in maize yield was not observed between the two treatments in 2019 (Figure 8). This result probably indicated that soil compaction may be ameliorated slightly in 2 yr due to the high shrinkage and swelling capacity of Vertisol and intense wetting and drying cycles (Figure 1). Pillai and McGarry (1999) verified that repeated wetting and drying cycles were the key processes in repairing degraded structures in Vertisols. The smaller difference in the penetration resistance between the compacted treatment and noncompacted treatment in 2019 compared with 2018 in the soil layer depth of 0–20 cm also supported this inference (Figure 2). The maize yield had a negative and significant correlation with the bulk density in the soil layer depth of 20–30 cm and penetration resistance in the soil layer depth of 30–50 cm in 2018 rather than in 2019 (Supplemental Table S4), indicating that soil compaction impeded maize yield in 2018 but not

in 2019. Another explanation was that soil compaction likely exerted a more adverse effect in wet years (2018) than in dry years (2019). Lipiec et al. (2003) also found that the sugar beet root yield was less influenced by soil compaction in the dry season than in the wet season. Xiong et al. (2020) revealed that soil compaction exerted a negative effect on maize root growth in wet soil conditions but not in dry soil conditions. On the one hand, soil compaction considerably decreased water infiltration and subsequently increased the potential risk of waterlogging in a wet year (Chyba et al., 2014). On the other hand, high soil moisture further worsened unfavorable soil aeration in compacted soil (Schjønning et al., 2013). Thus, the combined effects of soil compaction and waterlogging were more harmful for crop growth than soil compaction alone (Wu et al., 2018). Furthermore, the positive and significant correlation between the SWC in the soil layer depth of 0–80 cm and maize yield in 2019 (Supplemental Table S4) presumably suggested that the soil compaction effect in the dry season was masked by the soil water deficit (Lipiec et al., 2003). This result implies that the effect of soil compaction on maize yield is highly dependent on weather conditions.

In the current study, we found that the Rhv treatment in 2018 and the Rap treatment in 2019 significantly increased the maize yield compared with the Con treatment in the compacted soil conditions, but these treatments did not have the same effect in the noncompacted soil conditions, which coincided with the response of the root characteristics to cover cropping under different soil compaction levels (Figures 6, 7, and 8). This result was similar to that of Chen and Weil (2011), who reported that the maize silage yield was significantly enhanced by cover crops. They ascribed the cover crop effects on maize yields mainly to increased soil water availability to maize because the cover crop root channels, which provided pathways for maize roots to penetrate compacted soil layers to explore subsoil water. This explanation was likely applicable for our study in 2019, when the soil water deficit was a constraint on maize growth (Figure 1; Supplemental Figure S1; Supplemental Table S4). However, root channels created by cover crops may have benefited maize growth in 2018 by increasing water drainage and soil aeration because waterlogging frequently occurred in this year. Some previous studies also suggested that certain cover crops can increase the yield of the following crop by enhancing soil total N content (e.g., by N-fixing legumes or residue inputs (Alvarez et al., 2017; Blanco-Canqui et al., 2012; Tonitto et al., 2006); however, this effect decreased with an increasing rate of inorganic N application (Blanco-Canqui et al., 2012; Miguez & Bollero, 2005). Blanco-Canqui et al. (2012) and Miguez and Bollero (2005) reported no effect of cover crops on the succeeding crop yield when the rate of inorganic N fertilization was greater than 66 kg ha⁻¹ and 200 kg ha⁻¹, respectively. In this study, cover crop residues were removed and 180 kg ha⁻¹ inorganic N fertilizer was applied in the maize growing

season. Thus, we inferred that the cover crop effects on maize yield associated with soil nutrients were minor in our study.

The Rhv treatment had the greatest root biomass density and root length density in 2019 and maize yield in 2018 in the compacted plots (Figures 7 and 8). The result only partly supported our hypothesis that a mixture of radish and hairy vetch performed better than alfalfa and oilseed rape at ameliorating maize growth in compacted soil, because oilseed rape was better at improving root growth in 2018 and maize yield in 2019 in the compacted soil (Figures 7 and 8). Chen and Weil (2011) documented that both radish and rape improved maize root growth through compacted soil relative to no cover crop treatment. Although the Alf treatment was the most effective in improving the K_s and K_{a60} after 2 yr of cover cropping, the maize yield in the Alf treatment was the lowest relative to the Con, Rap, and Rhv treatments in 2019 (Figures 4, 5, and 8), which likely resulted from the lower SWC due to alfalfa regrowth and limited rainfall in the early stage of maize growth (Figure 1; Supplemental Figure S1) (Unger & Vigil, 1998). The variable effects of cover crop species on maize growth in compacted soil between the 2 yr with different weather conditions (Figures 7 and 8; Supplemental Figure S1) suggested that long-term studies are required. From the current 2 yr of experiment, we concluded that the cover crop treatments improved maize growth in the compacted soil in the order of Rap $\approx$ Rhv > Alf.

5 | CONCLUSIONS

Bio-tillage by cover crops decreased SWC along the soil profile during the cover crop growing season. Cover crops had no effect on the soil bulk density, and cover crops significantly altered the soil penetration resistance by affecting the SWC. The Alf treatment, but not the Rap and Rhv treatments, increased the K_s and K_{a60} in the subsoil, which was more pronounced in the noncompacted soil than in the compacted soil relative to the Con treatment. However, compared with the Con treatment, the cover crops improved the maize root characteristics and yield in the compacted plots but not in the noncompacted plots, suggesting that the root channels of the cover crops might have a considerable effect on maize growth in compacted soil. Generally, the Rap and Rhv treatments led to greater improvements in root characteristics and yield in maize than the Alf treatment under compacted conditions. The study demonstrated that bio-tillage benefits maize growth more prominently in compacted than in noncompacted soil. Bio-tillage can be used as an effective measure to improve air and water conductivities and maize growth in compacted soil.

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AUTHOR CONTRIBUTIONS

Zhongbin Zhang: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Writing – original draft; Writing – review & editing. Lei Yan: Data curation; Formal analysis; Investigation; Methodology. Yuekai Wang: Investigation; Writing – review & editing. Renjie Ruan: Investigation; Writing – review & editing. Peng Xiong: Investigation; Writing – review & editing. Xinhua Peng: Funding acquisition; Project administration; Resources; Writing – review & editing.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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